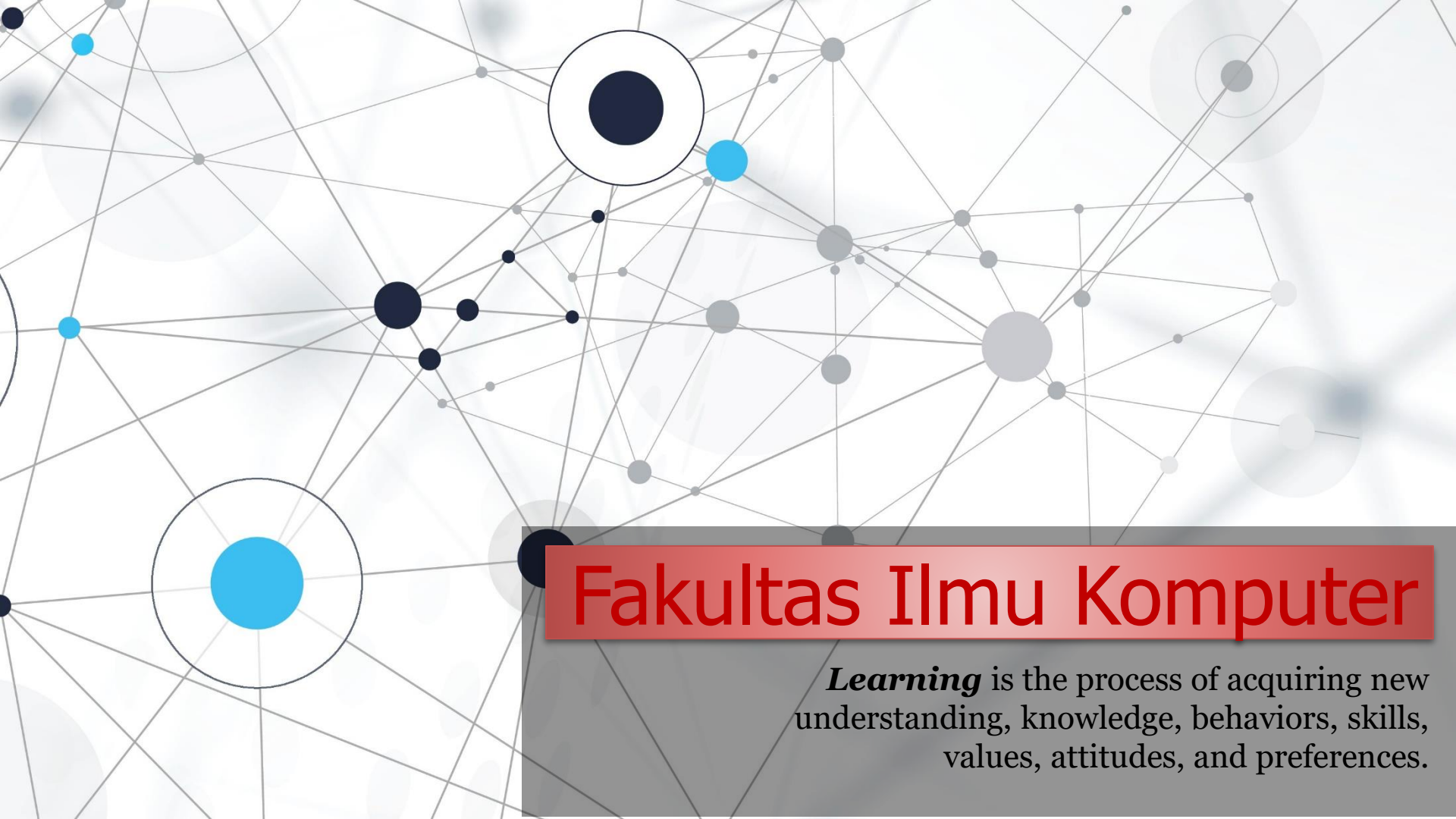


A complex network diagram with numerous nodes of varying sizes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger concentric circles. The background is white with faint grey circular patterns.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Framework #7-2

[Model & Database Query]

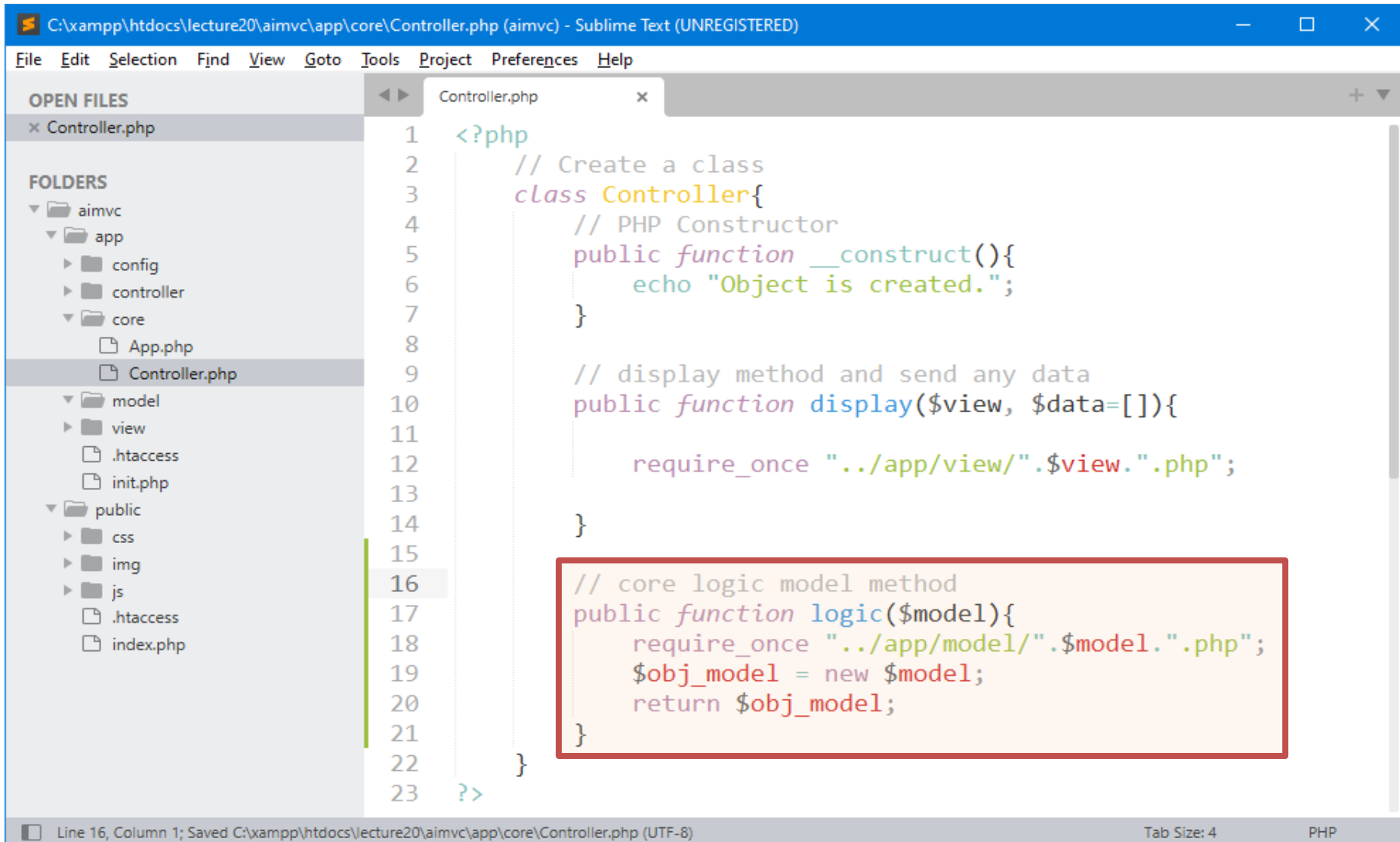


Welcome to the Framework Programming *course*

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some brownish edges, suggesting a natural, organic theme.

1 – Create Core Controller Class

1 - Create Core Logic Method



```
C:\xampp\htdocs\lecture20\aimvc\app\core\Controller.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Controller.php

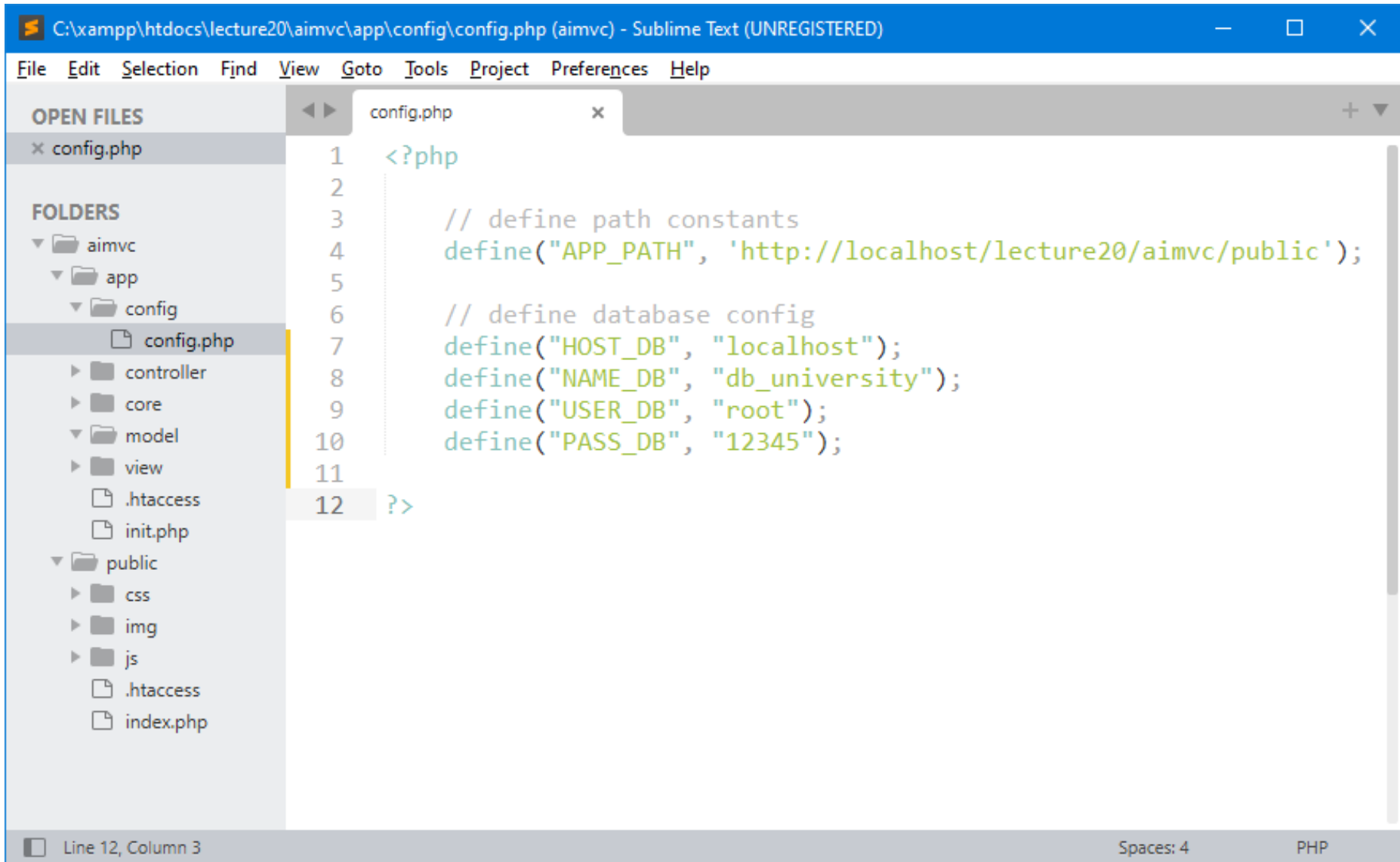
FOLDERS
aimvc
  app
    config
    controller
    core
      App.php
      Controller.php
    model
    view
      .htaccess
      init.php
  public
    css
    img
    js
    .htaccess
    index.php

1  <?php
2  // Create a class
3  class Controller{
4      // PHP Constructor
5      public function __construct(){
6          echo "Object is created.";
7      }
8
9      // display method and send any data
10     public function display($view, $data=[]){
11
12         require_once "../app/view/".$view.".php";
13
14     }
15
16     // core logic model method
17     public function logic($model){
18         require_once "../app/model/".$model.".php";
19         $obj_model = new $model;
20         return $obj_model;
21     }
22 }
23 ?>
```

Line 16, Column 1; Saved C:\xampp\htdocs\lecture20\aimvc\app\core\Controller.php (UTF-8) Tab Size: 4 PHP

Model component stores data and its ***related logic***.

2 - Config a Connection to MySQL



The screenshot shows the Sublime Text editor interface. The title bar indicates the file path is `C:\xampp\htdocs\lecture20\aimvc\app\config\config.php` and the editor is running on a Windows system. The menu bar includes File, Edit, Selection, Find, View, Goto, Tools, Project, Preferences, and Help. The left sidebar displays the 'FOLDERS' view, showing a tree structure of the project: `aimvc` (folder), `app` (folder), `config` (folder, selected), `controller` (folder), `core` (folder), `model` (folder), `view` (folder), `.htaccess` (file), `init.php` (file), `public` (folder), `css` (folder), `img` (folder), `js` (folder), `.htaccess` (file), and `index.php` (file). The main editor area shows the content of `config.php` with line numbers 1 through 12. The code defines path constants and database configuration. The status bar at the bottom indicates 'Line 12, Column 3', 'Spaces: 4', and 'PHP'.

```
1  <?php
2
3      // define path constants
4      define("APP_PATH", 'http://localhost/lecture20/aimvc/public');
5
6      // define database config
7      define("HOST_DB", "localhost");
8      define("NAME_DB", "db_university");
9      define("USER_DB", "root");
10     define("PASS_DB", "12345");
11
12  ?>
```

3 - Create Code Database Connection

```
C:\xampp\htdocs\lecture20\aimvc\app\core\Database.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

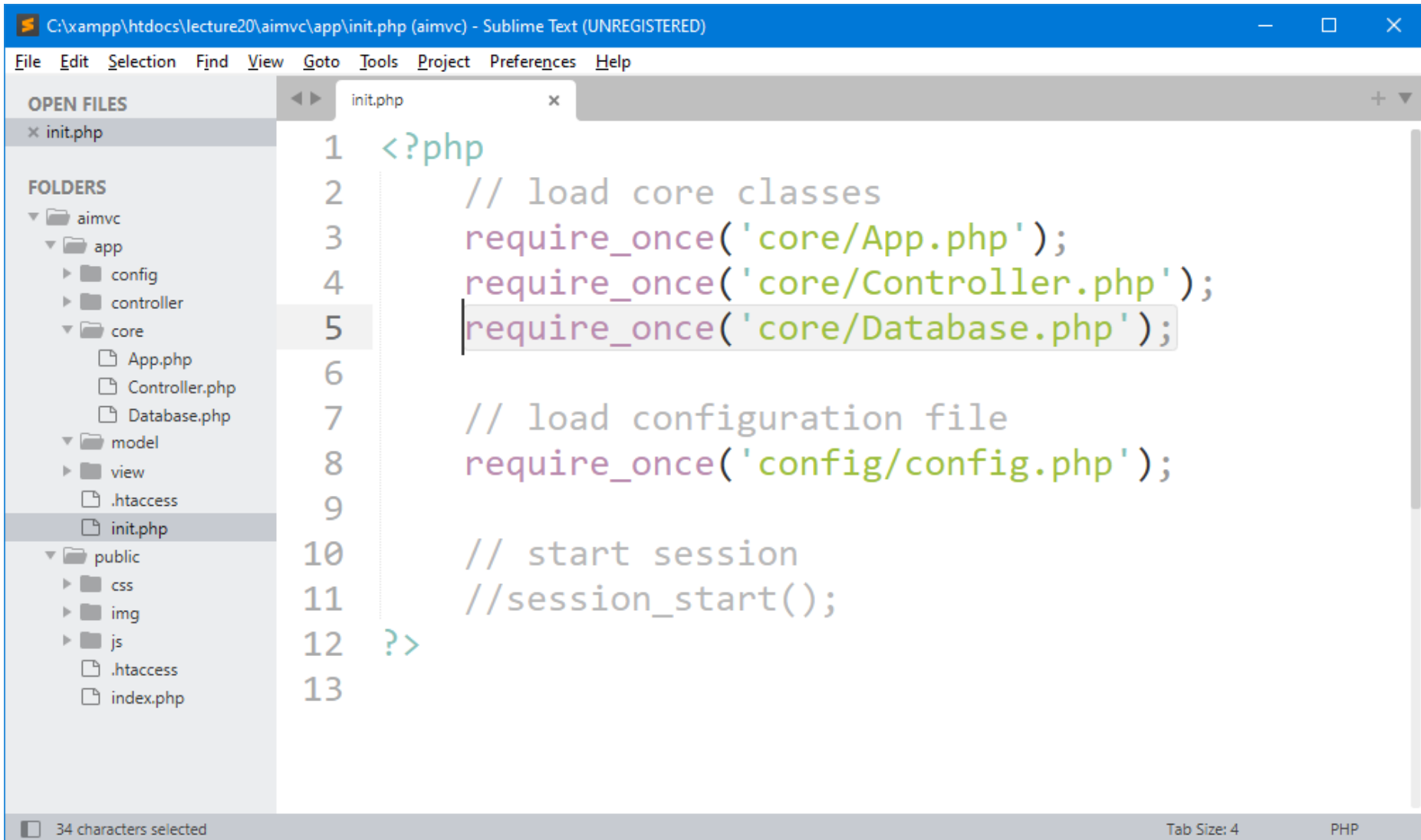
OPEN FILES
Database.php
Database.php

FOLDERS
aimvc
├── app
│   ├── config
│   ├── controller
│   └── core
│       ├── App.php
│       ├── Controller.php
│       └── Database.php
│   ├── model
│   ├── view
│   ├── .htaccess
│   └── init.php
└── public
    ├── css
    ├── img
    ├── js
    ├── .htaccess
    └── index.php

1 <?php
2 class Database{
3     private $server_name = HOST_DB;
4     private $db_name     = NAME_DB;
5     private $user_name   = USER_DB;
6     private $password    = PASS_DB;
7
8     private $con;
9
10    public function __construct(){
11        // check database connection
12        $this->con = $this->db_connection($this->server_name, $this->user_name, $this->password, $this->db_name);
13        if($this->con==false){
14            // status connection failed.
15        }else{
16            // status connected successfully.
17        }
18    }
19
20    private function db_connection($srvr_nm, $usr_nm, $psswr, $db_nm){
21        // create connection
22        $conn = new mysqli($srvr_nm, $usr_nm, $psswr, $db_nm);
23        // check connection
24        if ($conn->connect_error) {
25            //die("Connection failed: " . $conn->connect_error);
26            return false;
27        }
28        return $conn;
29    }
30
31    public function query($sql){
32        $result = $this->con->query($sql);
33
34        if ($result->num_rows > 0) {
35            // convert to associative array
36            $rows = mysqli_fetch_all($result, MYSQLI_ASSOC);
37            return $rows;
38        } else {
39            return []; // Empty array
40        }
41    }
42
43    public function db_close(){
44        $this->con->close();
45    }
46 }
47 ?>
```

Line 1, Column 1 Tab Size: 4 PHP

4 – Manage Core *Database.php* in *init.php*



C:\xampp\htdocs\lecture20\aimvc\app\init.php (aimvc) - Sublime Text (UNREGISTERED)

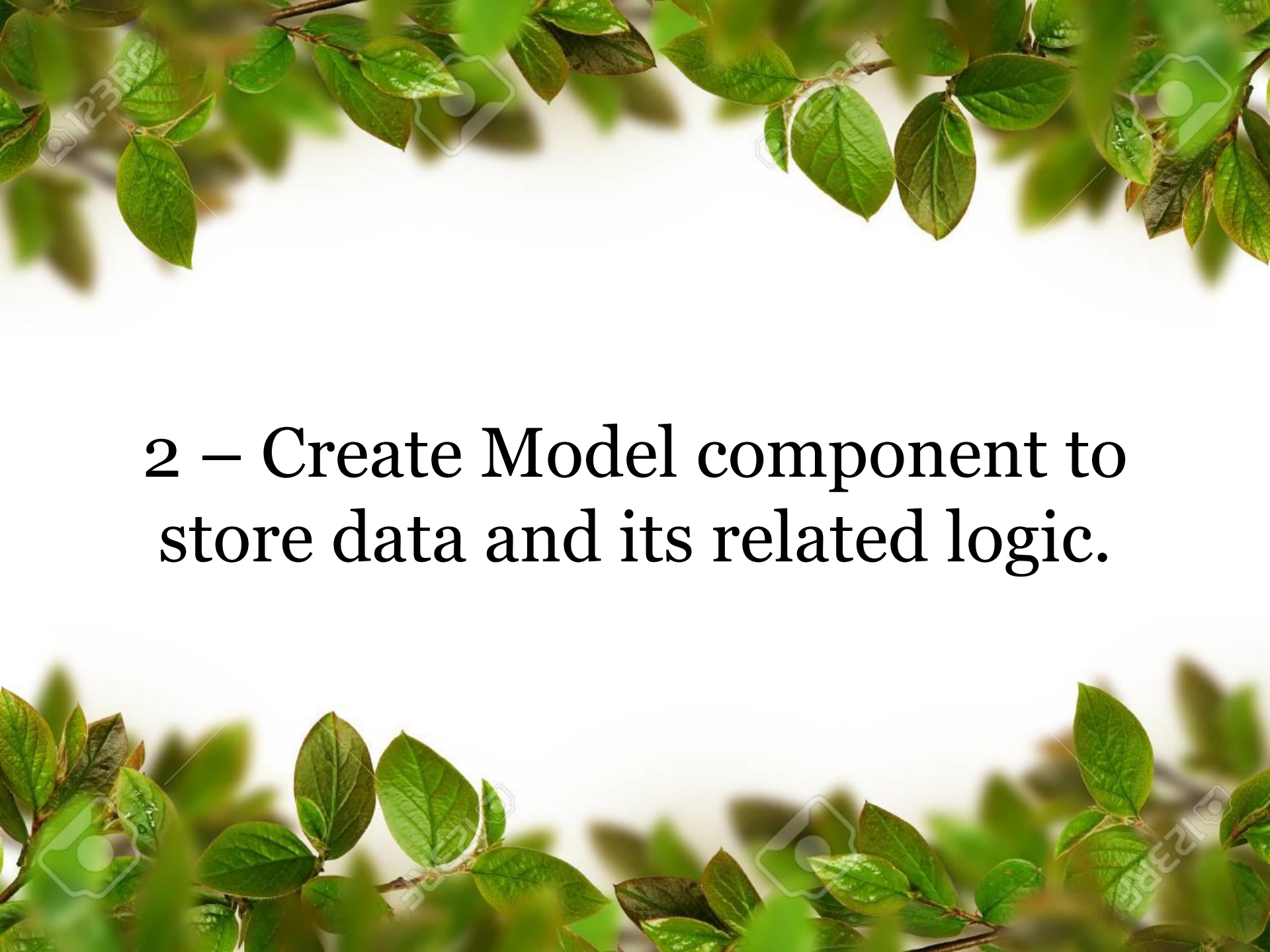
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x init.php

FOLDERS
▼ aimvc
▼ app
 config
 controller
 core
 App.php
 Controller.php
 Database.php
 model
 view
 .htaccess
 init.php
▼ public
 css
 img
 js
 .htaccess
 index.php

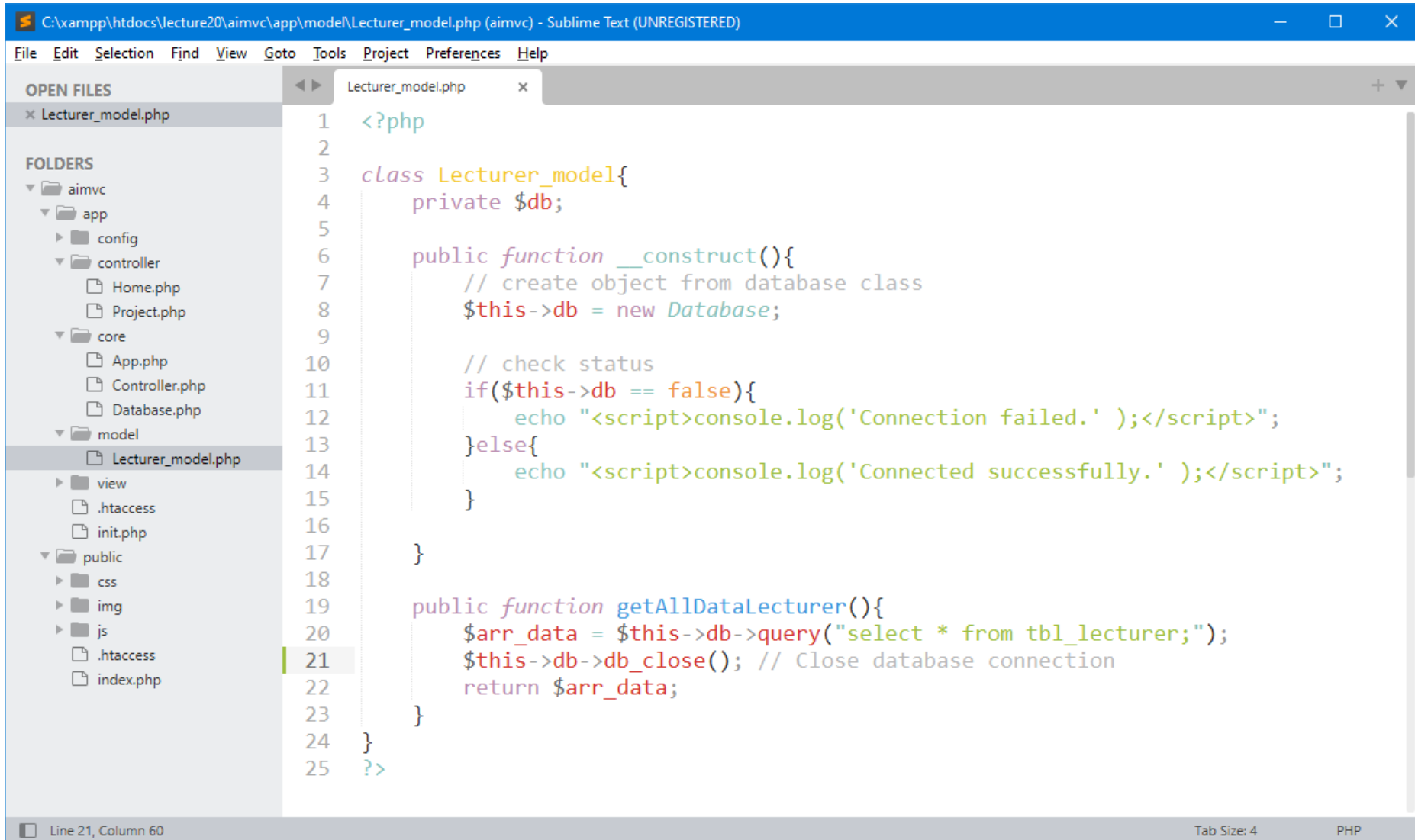
```
1  <?php
2      // load core classes
3      require_once('core/App.php');
4      require_once('core/Controller.php');
5      require_once('core/Database.php');
6
7      // load configuration file
8      require_once('config/config.php');
9
10     // start session
11     //session_start();
12  ?>
13
```

34 characters selected Tab Size: 4 PHP

The image features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in a black, serif font.

2 – Create Model component to store data and its related logic.

1 – Create Model Component (*Lecturer_model*)



```
C:\xampp\htdocs\lecture20\aimvc\app\model\Lecturer_model.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x Lecturer_model.php

FOLDERS
aimvc
  app
    config
    controller
      Home.php
      Project.php
    core
      App.php
      Controller.php
      Database.php
    model
      Lecturer_model.php
    view
      .htaccess
      init.php
  public
    css
    img
    js
    .htaccess
    index.php

1  <?php
2
3  class Lecturer_model{
4      private $db;
5
6      public function __construct(){
7          // create object from database class
8          $this->db = new Database;
9
10         // check status
11         if($this->db == false){
12             echo "<script>console.log('Connection failed.' );</script>";
13         }else{
14             echo "<script>console.log('Connected successfully.' );</script>";
15         }
16     }
17
18     public function getAllDataLecturer(){
19         $arr_data = $this->db->query("select * from tbl_lecturer;");
20         $this->db->db_close(); // Close database connection
21         return $arr_data;
22     }
23 }
24 }
25 ?>
```

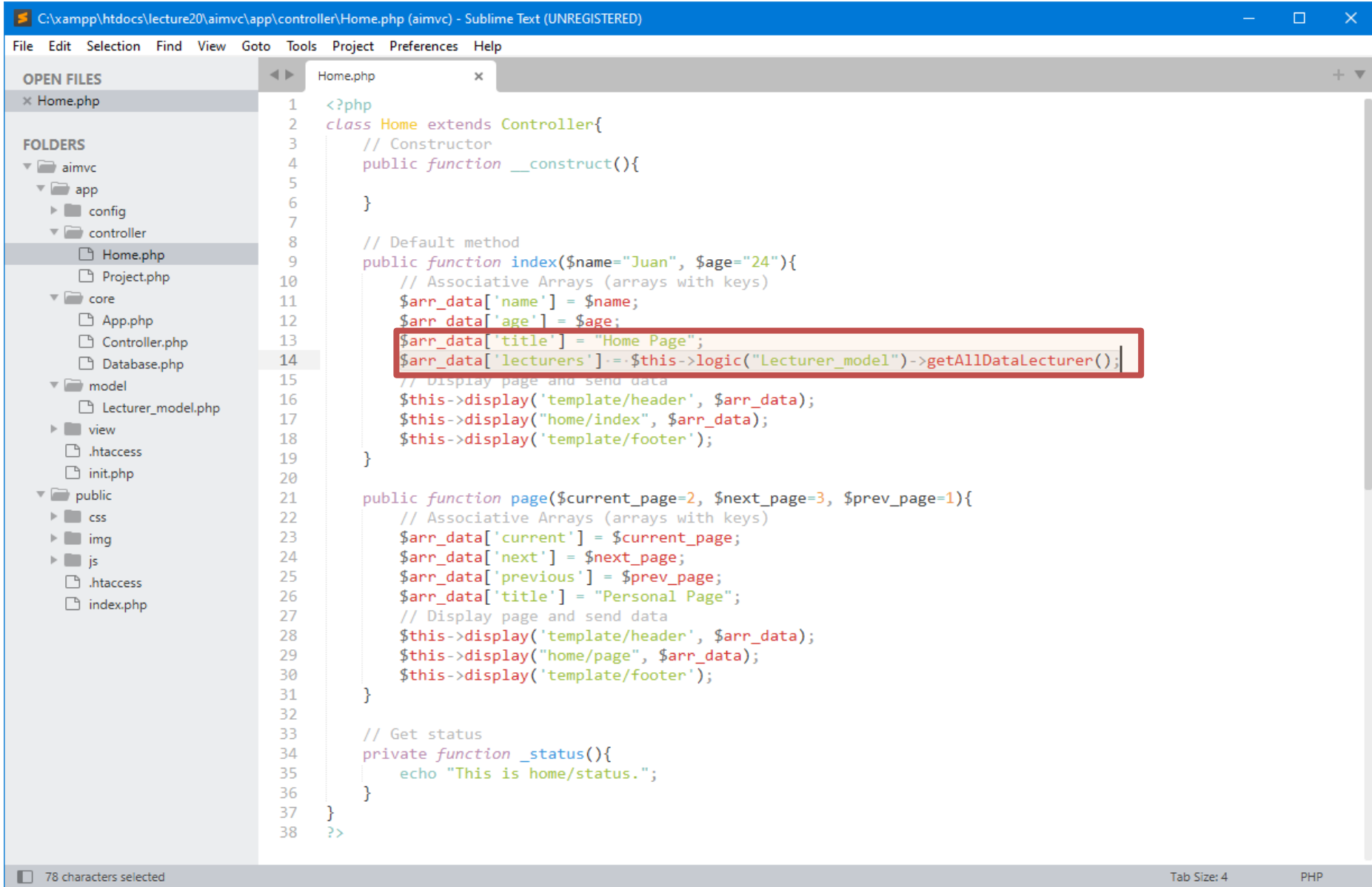
Line 21, Column 60

Tab Size: 4

PHP

Create model component with ***related logic*** and get the data from database.

2 – How to Read the *data* from Controller?

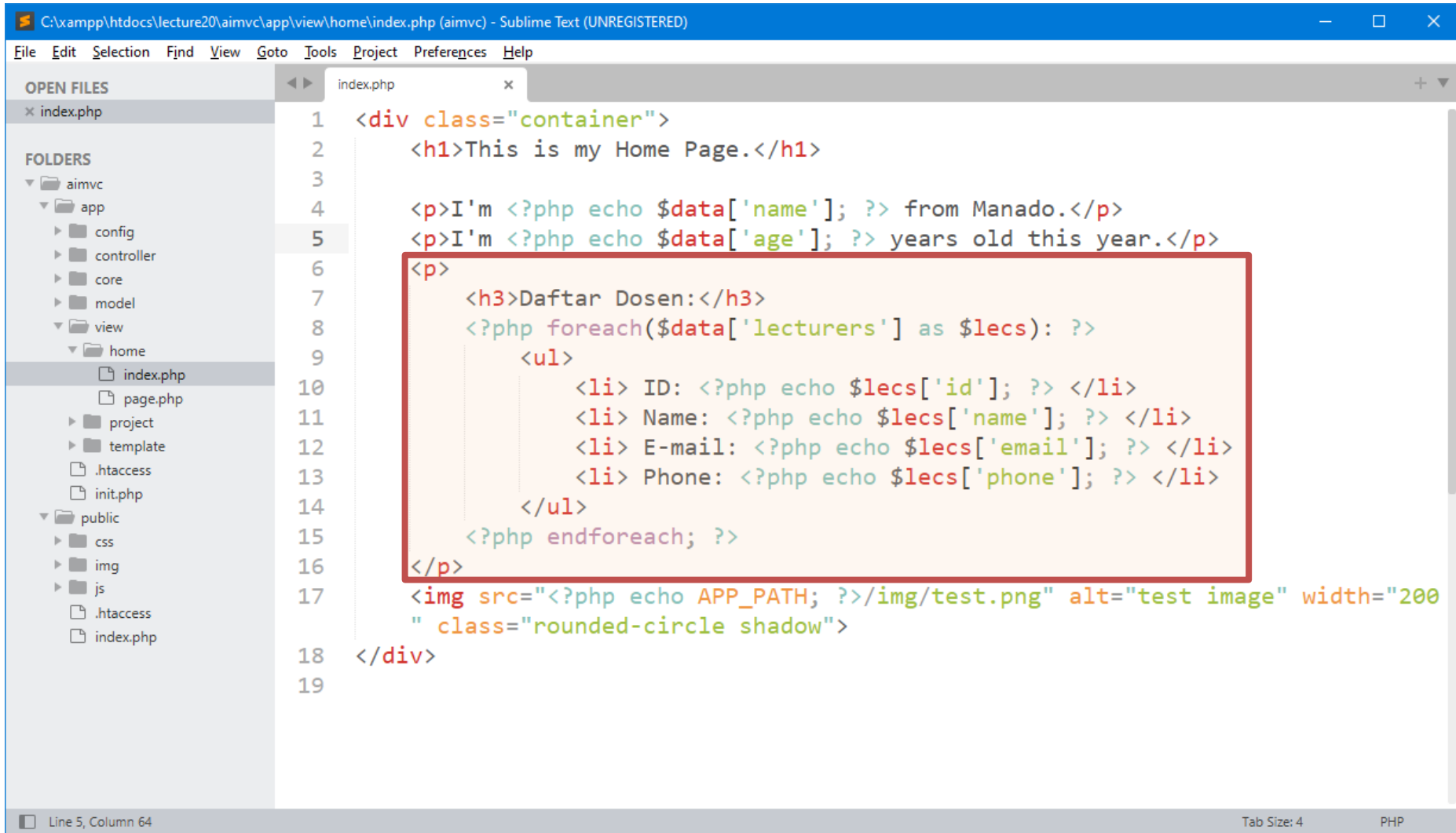


The screenshot shows a Sublime Text editor window titled "C:\xampp\htdocs\lecture20\aimvc\app\controller\Home.php (aimvc) - Sublime Text (UNREGISTERED)". The editor displays the following PHP code:

```
1 <?php
2 class Home extends Controller{
3     // Constructor
4     public function __construct(){
5
6     }
7
8     // Default method
9     public function index($name="Juan", $age="24"){
10         // Associative Arrays (arrays with keys)
11         $arr_data['name'] = $name;
12         $arr_data['age'] = $age;
13         $arr_data['title'] = "Home Page";
14         $arr_data['lecturers'] = $this->logic("Lecturer_model")->getAllDataLecturer();
15         // Display page and send data
16         $this->display('template/header', $arr_data);
17         $this->display("home/index", $arr_data);
18         $this->display('template/footer');
19     }
20
21     public function page($current_page=2, $next_page=3, $prev_page=1){
22         // Associative Arrays (arrays with keys)
23         $arr_data['current'] = $current_page;
24         $arr_data['next'] = $next_page;
25         $arr_data['previous'] = $prev_page;
26         $arr_data['title'] = "Personal Page";
27         // Display page and send data
28         $this->display('template/header', $arr_data);
29         $this->display("home/page", $arr_data);
30         $this->display('template/footer');
31     }
32
33     // Get status
34     private function _status(){
35         echo "This is home/status.";
36     }
37 }
38 ?>
```

The left sidebar shows the project structure with folders like aimvc, app, controller, core, model, view, and public. The file Home.php is selected in the controller folder. The status bar at the bottom indicates "78 characters selected", "Tab Size: 4", and "PHP".

3 – Modify Code in /view/home/index.php



```
C:\xampp\htdocs\lecture20\aimvc\app\view\home\index.php (aimvc) - Sublime Text (UNREGISTERED)
File Edit Selection Find View Goto Tools Project Preferences Help

OPEN FILES
x index.php

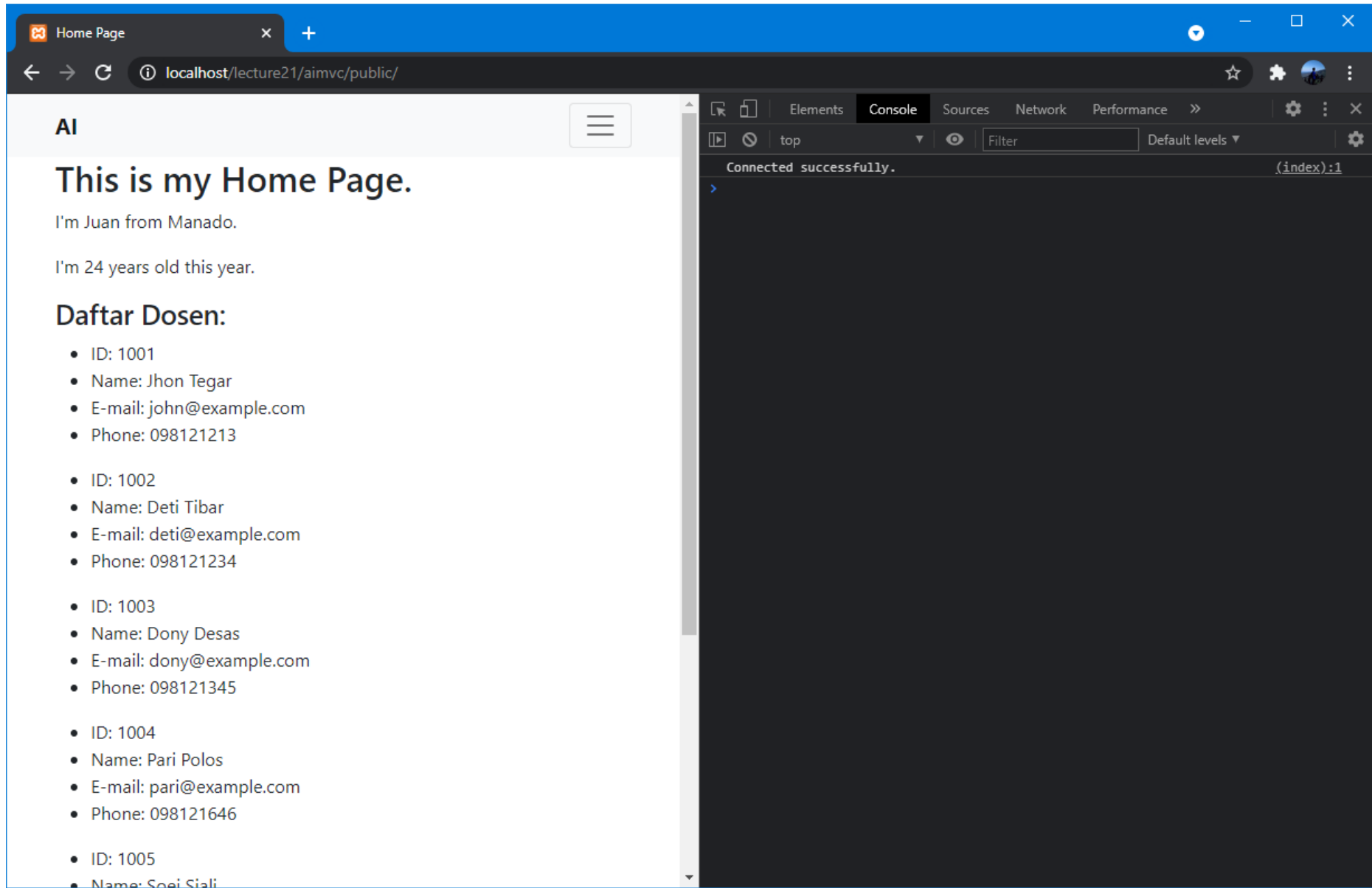
FOLDERS
aimvc
├── app
│   ├── config
│   ├── controller
│   ├── core
│   ├── model
│   └── view
│       ├── home
│       │   ├── index.php
│       │   └── page.php
│       ├── project
│       └── template
├── .htaccess
├── init.php
├── public
│   ├── css
│   ├── img
│   ├── js
│   ├── .htaccess
│   └── index.php

1 <div class="container">
2     <h1>This is my Home Page.</h1>
3
4     <p>I'm <?php echo $data['name']; ?> from Manado.</p>
5     <p>I'm <?php echo $data['age']; ?> years old this year.</p>
6     <p>
7         <h3>Daftar Dosen:</h3>
8         <?php foreach($data['lecturers'] as $lecs): ?>
9             <ul>
10                 <li> ID: <?php echo $lecs['id']; ?> </li>
11                 <li> Name: <?php echo $lecs['name']; ?> </li>
12                 <li> E-mail: <?php echo $lecs['email']; ?> </li>
13                 <li> Phone: <?php echo $lecs['phone']; ?> </li>
14             </ul>
15         <?php endforeach; ?>
16     </p>
17     
19 </div>
```

Line 5, Column 64

Tab Size: 4 PHP

4 – Testing View Data *from* Database



The screenshot shows a web browser window with a single tab titled "Home Page". The address bar displays "localhost/lecture21/aimvc/public/". The page content includes a heading "AI", a main heading "This is my Home Page.", and two paragraphs: "I'm Juan from Manado." and "I'm 24 years old this year." Below these is a section titled "Daftar Dosen:" followed by a list of five faculty members, each with their ID, Name, E-mail, and Phone number.

The Chrome DevTools console is open on the right side of the browser window. The "Console" tab is selected, showing a message "Connected successfully." with a source of "(index):1". The console also displays a blue arrow cursor at the top of the message area.

AI

This is my Home Page.

I'm Juan from Manado.

I'm 24 years old this year.

Daftar Dosen:

- ID: 1001
- Name: Jhon Tegar
- E-mail: john@example.com
- Phone: 098121213

- ID: 1002
- Name: Deti Tibar
- E-mail: deti@example.com
- Phone: 098121234

- ID: 1003
- Name: Dony Desas
- E-mail: dony@example.com
- Phone: 098121345

- ID: 1004
- Name: Pari Polos
- E-mail: pari@example.com
- Phone: 098121646

- ID: 1005
- Name: Soei Siali



exercise for students

Latihan untuk *Students*

Create directory : C:\xampp\htdocs\lecture21\
Create file : `_.php`

Buat semua contoh yang sudah di pelajari sebelumnya di *lecture* ini.

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU



References

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